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Ventures

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Report for the Final Project

I began by making two fundamental assumptions that my work would be based upon. The first was that each pair of numbers represented one letter. The second was that the spaces in the cipher had the significance of spacing out words. This meant that a pair of numbers that were isolated using spaces represented a single letter. Since I knew the message was in English from the fact that the message was recovered from England, I focused on pairs of numbers that appeared by themselves. The only letters that appear by themselves as words in the English language are “A” and “I”. This meant each pair of numbers that was by itself was either “A” or “I”. I wrote down each pair of numbers that appeared by itself and found that they were much more than two. Which meant that “A” and “I” were represented by multiple numbers. I then went on to check the numbers that appeared in the cipher and noticed that all numbers from 0 to 99 appeared in the cipher at least once with exception of 16. These discoveries led me to believe that the cipher I was dealing with was homophonic since the homophonic cipher gives symbols to letters depending on their percentage of frequency and having 99 numbers out of a possible 100 representing letters was a strong sign.

Upon making this assumption, I carried out frequency analysis on the numbers. One thing I immediately noticed was that the number “19” appeared twice in the cipher, which was a bit far off the amount each number is supposed to appear which is about 4.6 times for this cipher (each symbol in a homophonic cipher is supposed to appear about 1% of the time which is about 4.6 times in this case). The interesting thing with 19, was that the two times it

appeared was when they were right next to each other. This explanation for this was that there was no alternative when encrypting since the same number was used to encrypt the same letter when it repeated next to it. This meant that 19 represented a letter that only had one number to represent it. I looked at the example of the chart of a homophonic cipher from the code book and noticed that the letters that were represented by a single number were J,K,Q,V,X and Z [1]. From the position of the 19 on the word, I believed that 19 represented Z.

The next thing I did was that I looked at other numbers that were outliers in terms of frequency, the ones that came to mind quickly were 79 and 31. They each respectively appeared 8 and 7 times and the reason I paid attention to them specifically was that they both started three letter words and appeared frequently at the end of letters. The only doubt I had because of them was that 31 appeared at the end of two letter words twice while 79 did not. This led me to my most important discovery of the process as to explain this, I tried to see if the person who was encrypting the message alternated between the two letters when encoding. Upon checking the cipher, I saw that the letters alternated perfectly throughout the cipher. This meant that 31 and 79 were representing a single letter that is usually rare since it only had two numbers to represent it. From this discovery, I checked the code book again, for letters that were represented by 2 symbols and saw that Y was among these letters which would make perfect sense, as this symbol appeared in front and at the end of three letter words [1]. I concluded that 31 and 79 both represented Y.

This consequently meant that 82, the number that was behind 31 when they formed two letter words, was M. In addition to this, I made the assumption that the second number in the three letter words starting with either 31 or 79 represented O, this included numbers such as 00, 73, 92 and 88 which proved to be the true in the end.

Another outlier that was present was the number 25 which appeared 9 times which was much higher than how many times it should have appeared and it also behaved like no other letter as it started long letters and also ended smaller letters. I believed that 25 represented a letter which was represented by a single number. Since Z was taken it meant that 25 could only be one of J,K,Q,V and X, however I could not break it down further than that.

Since I had seen 31 and 79 alternate, I assumed that other numbers which represented the same letters would also alternate. I decided to test this assumption on the numbers that appeared solitarily. I was able to pick a certain group of 5 numbers which I believed to be the letter "I" because they had appeared at least once by themselves had formed a strong pattern alternating nearly perfectly with each other in the order of 39 – 35 – 61 – 86 – 30. The number 55 did not fit the pattern as well as others but its position at times made me also believe that it was also the letter "I". Since I had figured which numbers were I, I could begin figuring which numbers were N, F or T as these would be the numbers at the end of two letter words starting with the number I.

Another tactic I used to discover other letters was by finding similar word patterns. For example, I found a strand of very similar numbers which was 29 54 79| 65 02| 17 61 14| 68 87 46 71| 29 86 69 57| 99 26 76|25 62 88 12 93 36 96 78 15| together with 12 54 79| 65 02| 96 55 32| 68 26 46 62| 29 30 69 67| 99 23 50|25 70 88 12 93 14 80 78 76|. The numbers in these two strands were nearly identical which meant they were probably the same words being written. This meant that the numbers which were different at the same positions in the words represented the same letter being represented by different number. Through doing this I would be able to find strings of numbers that would represent a letter. Once I had a string of letters, that I knew represented the same thing I would try and identify which letter they represented.

For example, I found a strand for letters which I believed was E based on their position and other factors. The letter is supposed to be represented by about 12 symbols, but I only had about 6 letters. To confirm this, I checked if they appeared in consequent of each other, if a number did fit the sequence I would keep it, if it didn't I would remove it. Since I knew there were more numbers that needed to fit the sequence which represented the letter E, I looked for numbers in the cipher which fitted the sequence at each exact point, if they did fit the sequence I would go on to assume that they also represented the letter E. The following page contains the pattern I formed for the letter E where it can be seen that each of the letters appeared in a certain pattern each time. The risk with this was that some numbers could have appeared by coincidence, and these represented errors afterwards however most of the numbers that I found using this method were correct. The pattern I had for the letter E was 14 – 76 – 36 – 15 – 98 – 18 – 81 – 51 – 47 – 32 – 50, which repeated over and over in the text in the same order. The weakness of this was that if a number truly represented the letter and didn't fall in this pattern, I would disregard it at first but I would find it eventually.

Another example of using strands would be how I found the numbers that were the true identity of "A" where I used the numbers that had appeared solitarily but were not "I" to form strands of numbers that I believed to be A and used patterns to confirm the numbers I had. Throughout of this I also did a lot of trial and error where I made calculated guesses of what numbers could represent letters such as T, R and S based on strands that I had obtained and tried them, out where sometimes the guess was successful and at times it wasn't.

All in all, I used the patterns of numbers that represented particular letters to confirm whether my suspicion of a letter being represented by a particular number was right and as a result was able to form a list of numbers that represented different letters. Once I felt I had discovered the true identity of more than half the numbers in the cipher, I plugged their letter forms in the cipher and tried to piece out the true identity of letters through the holes I had in

found in the words, if a particular assumption worked out I went on with it, but if it didn't work I would try a new assumption of what the word was. I was also able to put letters to numbers after plugging in what I knew, such as finding out that the letter identity of 25 was "K" and that the letter identity of 27 which had only appeared once in the cipher was "J".

Decrypted Message

YOU ARE NOT QUITE FINISHED WITH THIS COURSE. I HOPE YOU ENJOYED THE QUIZZES AND OTHER FUN THINGS WE DID IN HERE. I HAVE QUICKLY BECOME FOND OF ALL OF YOU AND I HOPE YOU HAVE A HAPPY HOLIDAY BREAK. THANK YOU FOR PUTTING UP WITH MY QUIRKS. HERE IS ONE LAST PASSAGE. I CAN'T THINK OF A SADDER WAY TO DIE THAN WITH THE KNOWLEDGE THAT I NEVER SHOWED UP IN THIS WORLD AT WHO I REALLY AM. I CAN'T THINK OF A MORE GRACED WAY TO DIE THAN WITH THE KNOWLEDGE THAT AT BEST I KNEW HOW I SHOWED UP HERE, AS MY TRUE SELF, ABLE TO ENGAGE THE WORLD IN FREEDOM AND WITH LOVE BECAUSE I HAD BECOME FIERCE WITH REALITY.

References

[1] Singh, Simon. *The Code Book: The Science of Secrecy from Ancient Egypt to Quantum Cryptography,* Anchor, 2000